Infosys Springboard Virtual Internship Program

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Starting Date: 29/12/2025

Duration: 2 months

Design and Implementation of a Multi-Model AI Agent for Automated Health Diagnostics

Internship Technical Report – Reformatted (Architecture-Based Structure)

1. Introduction

This project focuses on the design and implementation of a modular multi-model AI system for automated interpretation of laboratory blood reports.

Blood reports often contain clinically significant parameters such as Hemoglobin, RBC, WBC, Platelets, MCV, RDW, and Uric Acid. However, interpreting these reports requires medical expertise and contextual reasoning.

The objective of this system is to:

- Automatically ingest reports from multiple formats
- Extract structured clinical parameters
- Interpret individual values against reference ranges
- Detect clinically relevant patterns
- Assess risk levels
- Generate structured, personalized health recommendations
- Provide a complete end-to-end diagnostic support pipeline

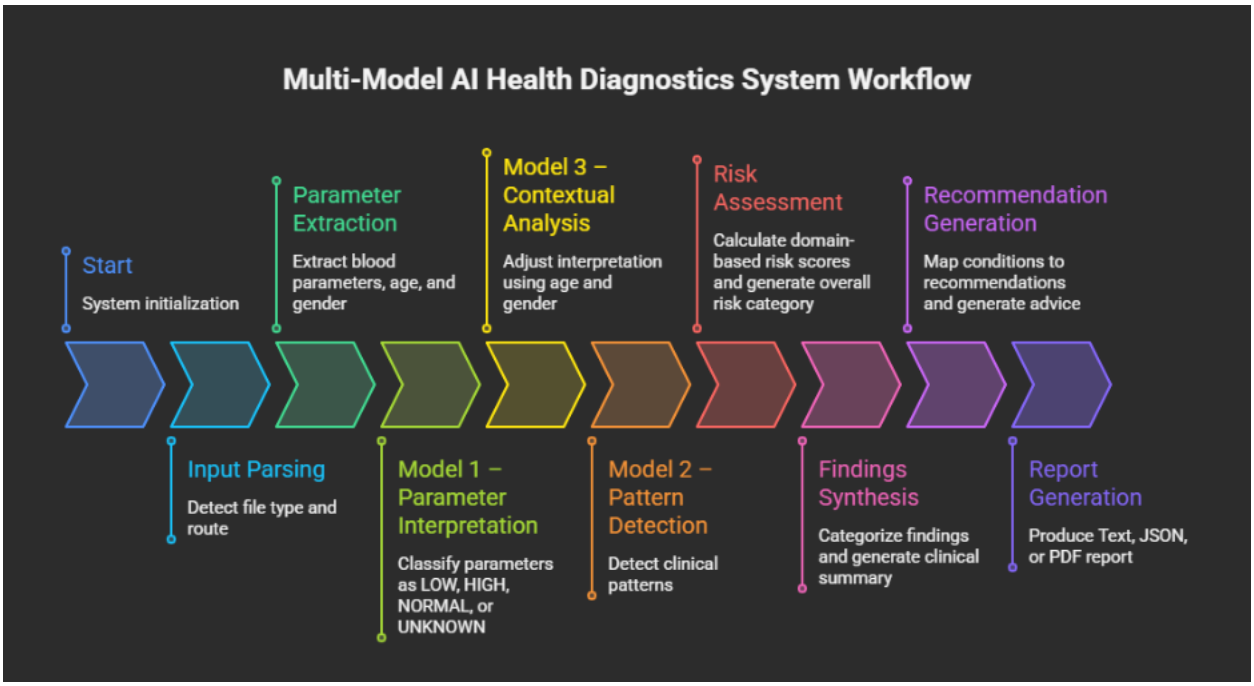
2. System Architecture Overview

The system follows a modular multi-model architecture composed of:

- Input Interface & Parser
- Data Extraction Engine
- Reference Range Management Module
- Model 1 – Parameter Interpretation
- Model 3 – Contextual Analysis

- Model 2 – Pattern Detection & Risk Assessment
- Findings Synthesis Engine
- Recommendation Generator
- Report Generation Module
- Multi-Model Orchestrator

Each module performs a clearly defined responsibility, ensuring scalability, modularity, and fault tolerance.



3. Data Ingestion & Multi-Format Input Handling

The system supports:

- JSON reports
- PDF reports
- Image-based reports (via OCR)

A file-type detection mechanism routes inputs to appropriate parsers.

For PDF and image-based reports, OCR-based extraction is applied where necessary.

This enables real-world extensibility, where lab reports may vary in format and structure.

4. Data Extraction Engine

The extraction module identifies:

- Blood parameters (Hemoglobin, RBC, WBC, Platelets, etc.)
- Corresponding numeric values
- Units
- Reference ranges (if present)
- Demographic information (Age and Gender)

Extraction mechanisms include:

- Keyword-based parameter mapping
- Regular expression pattern matching
- Multiple fallback patterns for demographic fields

Age extraction supports formats such as:

- “Age: 25 years”
- “Age/Gender: 25/Male”
- Standalone demographic formats

OCR inconsistencies are handled using fallback parsing strategies.

5. Reference Range Management

A centralized reference range loader was implemented using a configuration-based JSON structure.

It supports:

- Age-specific adjustments

- Gender-specific adjustments
- Fallback to general reference ranges

This ensures:

- Flexibility
- Maintainability
- Scalability for future clinical expansion

Reference ranges are converted into numeric thresholds before evaluation.

```
"parameters": {
    "Hemoglobin": {
        "unit": "g/dl",
        "description": "Oxygen-carrying protein in red blood cells",
        "general": {"min": 12.0, "max": 17.0},
        "male": {
            "neonate": {"min": 14.0, "max": 24.0},
            "infant": {"min": 10.0, "max": 15.0},
            "child": {"min": 11.5, "max": 14.5},
            "teenager": {"min": 13.0, "max": 16.0},
            "adult": {"min": 13.5, "max": 17.5},
            "senior": {"min": 12.5, "max": 17.0}
        },
        "female": {
            "neonate": {"min": 14.0, "max": 24.0},
            "infant": {"min": 10.0, "max": 15.0},
            "child": {"min": 11.5, "max": 14.5},
            "teenager": {"min": 12.0, "max": 15.0},
            "adult": {"min": 12.0, "max": 15.5},
            "senior": {"min": 11.5, "max": 15.0},
            "pregnant": {"min": 11.0, "max": 14.0}
        }
    },
```

6. Model 1 – Parameter Interpretation

Each extracted parameter is classified into:

- LOW
- HIGH
- NORMAL
- UNKNOWN (if insufficient data)

Classification is performed using boundary comparison against reference thresholds.

The system supports multiple reference range formats and converts them into numeric evaluation logic.

Missing parameters are handled gracefully without pipeline failure.

This model forms the foundation of the interpretation layer.

DETAILED PARAMETERS

Parameter	Value	Range	Status
Hemoglobin	14.5 g/dl	13.0-16.5	NORMAL
RBC	4.79 10^6/uL	4.5-5.5	NORMAL
WBC	10.57 10^3/uL	4.0-10.0	HIGH
Platelet	150.0 10^3/uL	150.0-410.0	NORMAL
HCT	43.3 %	40.0-49.0	NORMAL
MCV	90.3 fl	83.0-101.0	NORMAL
MCH	30.2 pg	27.1-32.5	NORMAL
MCHC	33.4 g/dl	32.5-36.7	NORMAL
RDW-CV	13.6 %	11.6-14.0	NORMAL
Urea	18.0 mg/dl	19.3-43.0	LOW
Uric Acid	4.9 mg/dl	3.5-8.5	NORMAL

7. Model 3 – Contextual Analysis

Contextual refinement is performed before pattern detection.

Adjustments are based on:

- Age
- Gender

Examples:

- Hemoglobin reference ranges differ between males and females
- Age-based thresholds vary across life stages

Contextual refinement updates parameter status dynamically prior to correlation analysis.

This improves demographic accuracy and clinical reliability.

```
------------------------
 PATIENT INFORMATION
------------------------
 Age       : 65 years
 Gender    : Female
 Age Group: Senior
```

8. Model 2 – Pattern Detection

The system moves beyond isolated parameter interpretation and evaluates parameter correlations.

Implemented clinical pattern logic includes:

- Anemia indicators
- Kidney-related concerns
- Platelet abnormalities
- Infection indicators (WBC-based)
- RBC-only abnormality patterns
- MCV and RDW concerns
- Elevated uric acid pattern

Pattern detection logic:

- Counts abnormal indicators
- Applies severity-based deviation scoring
- Calculates confidence levels (capped to prevent overconfidence)
- Handles missing parameters safely

This enables correlation-based reasoning rather than isolated evaluation.

```
----------------------------------------------------------------------
  DETECTED PATTERNS
----------------------------------------------------------------------

  Pattern 1: Possible Infection/Inflammation
  Confidence: 51.4%
  Assessment: Elevated WBC may indicate infection or inflammation
  Indicators:
    - Elevated WBC: 10.57 10^3/uL (range: 4.0-10.0)

----------------------------------------------------------------------
  RISK ASSESSMENT
----------------------------------------------------------------------

  Overall Risk Score: 22.0/100
  Risk Level: LOW RISK

  Risk Breakdown:
    - Infection Risk: 30/100
        Elevated WBC: 10.57 10^3/uL (range: 4.0-10.0) (+30)
```

9. Risk Assessment Model

A domain-based risk scoring system aggregates contribution from:

- Anemia risk
- Kidney risk
- Platelet risk
- Infection risk

- RBC/MCV/RDW risk
- Uric acid risk

Each domain contributes to a weighted score based on severity.

The aggregated score is normalized to a 0–100 scale and categorized into:

- Minimal Risk
- Low Risk
- Moderate Risk
- High Risk
- Critical Risk

This provides a structured health summary rather than isolated alerts.

10. Findings Synthesis Engine

The synthesis module organizes findings into:

- Critical
- Abnormal
- Borderline
- Normal
- Not Found

Severity is calculated using deviation percentage from reference bounds.

The engine:

- Aggregates parameter classifications
- Integrates detected patterns
- Incorporates overall risk level
- Generates a clinical-style narrative summary

This ensures structured interpretation of output.

```
CLINICAL SUMMARY
Analysis for female patient, 65 years old (senior). Found 5 parameter(s) outside normal range. 1 parameter(s) at borderline levels. Identified conditions: Anemia Indicators, Microcytic Hypochromic Pattern. Overall risk assessment: LOW RISK (25.7/100).
```

11. Condition Identification

Detected patterns are translated into suspected conditions.

If abnormalities exist but no pattern is triggered:

- Isolated findings are handled independently

This ensures that no clinically relevant deviation is ignored.

12. Personalized Recommendation Generator

Recommendations are driven by a configuration-based JSON knowledge base.

Each recommendation includes:

- Diet suggestions
- Lifestyle modifications
- Follow-up advice
- Severity-based priority
- Age-specific notes
- Gender-specific considerations

Condition names are mapped to structured recommendation templates.

A priority aggregation mechanism ensures that the most severe condition determines advisory urgency.

Full traceability is maintained:

Parameter → Pattern → Condition → Recommendation

```
----------------------------------------------------------------------
  PERSONALIZED RECOMMENDATIONS
----------------------------------------------------------------------

  Priority Level: LOW
  Summary: Minor concerns identified: Anemia Indicators. Follow the dietary and lifestyle recommendations below. Retest in 4-6 weeks.

  ============================================================
  CONDITION: ANEMIA INDICATORS
  Timeline: 4-6 weeks follow-up
  ============================================================

  DIETARY RECOMMENDATIONS:
    1. Increase iron-rich foods: spinach, lentils, red meat, beans, fortified cereals
    2. Add Vitamin C sources with meals: oranges, tomatoes, bell peppers (improves iron absorption)
    3. Include Vitamin B12 sources: eggs, dairy, fish, fortified foods
    4. Add folate-rich foods: leafy greens, legumes, asparagus
    ... and 1 more recommendations

  LIFESTYLE MODIFICATIONS:
    1. Avoid tea or coffee with meals (tannins block iron absorption)
    2. Cook in cast iron pots (adds dietary iron)
    3. Space calcium supplements away from iron-rich meals
    4. Get adequate sleep (7-8 hours) for blood cell production
    ... and 1 more recommendations

  FOLLOW-UP ACTIONS:
    1. Retest CBC (Complete Blood Count) in 4-6 weeks
    2. Consider iron studies (serum ferritin, TIBC) if anemia persists
    3. Consult doctor if symptoms like fatigue, weakness, or dizziness occur
    4. Check for underlying causes if anemia doesn't improve with diet

  IMPORTANT WARNINGS:
    1. Do not take iron supplements without medical advice
    2. Excessive iron can be harmful - follow prescribed dosage only
```

13. Multi-Model Orchestrator

The Multi-Model Orchestrator executes the workflow sequentially:

1. Input Parsing
2. Parameter Extraction
3. Model 1 Interpretation
4. Model 3 Contextual Analysis
5. Model 2 Pattern Detection
6. Risk Assessment
7. Findings Synthesis
8. Recommendation Generation

Responsibilities include:

- Execution sequencing
- Data flow coordination

- Exception handling
- Status logging
- Graceful degradation

The pipeline does not crash due to partial data.

14. Report Generation Module

The system generates:

- Structured text reports
- JSON output reports
- PDF reports (via HTML-based rendering)

Reports include:

- Patient demographic summary
- Parameter overview
- Detected patterns
- Risk level
- Categorized findings
- Personalized recommendations
- Medical disclaimer

```
----------------------------------------------------------------------
  PATIENT-SPECIFIC NOTES
----------------------------------------------------------------------

  Age-Related Considerations:
  Gender-Related Considerations:
  Gender-Related Considerations:
    Lower hemoglobin ranges. Iron deficiency more common due to menstruation.   Gender-Related Considerations:
    Lower hemoglobin ranges. Iron deficiency more common due to menstruation. Monitor during pregnancy.

----------------------------------------------------------------------
  Gender-Related Considerations:
    Lower hemoglobin ranges. Iron deficiency more common due to menstruation. Monitor during pregnancy.

----------------------------------------------------------------------
  GENERAL HEALTH GUIDELINES
----------------------------------------------------------------------   Gender-Related Considerations:
    Lower hemoglobin ranges. Iron deficiency more common due to menstruation. Monitor during pregnancy.

----------------------------------------------------------------------
  GENERAL HEALTH GUIDELINES
----------------------------------------------------------------------

  Gender-Related Considerations:
    Lower hemoglobin ranges. Iron deficiency more common due to menstruation. Monitor during     4. Stress Management: Practice stress management: meditation, deep breathing, hobbies
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  4. Stress Management: Practice stress management: meditation, deep breathing, hobbies
  5. Regular Checkups: Regular annual health checkups are important

----------------------------------------------------------------------
  DISCLAIMER
----------------------------------------------------------------------

  This report is generated by an AI-based health diagnostics system
  for informational purposes only. It is NOT a substitute for
  professional medical advice, diagnosis, or treatment.

  Always consult a qualified healthcare provider for:
    - Interpretation of these results
    - Medical advice and treatment decisions
    - Any health concerns or symptoms

  Do not disregard professional medical advice or delay seeking it
  based on information in this report.
```
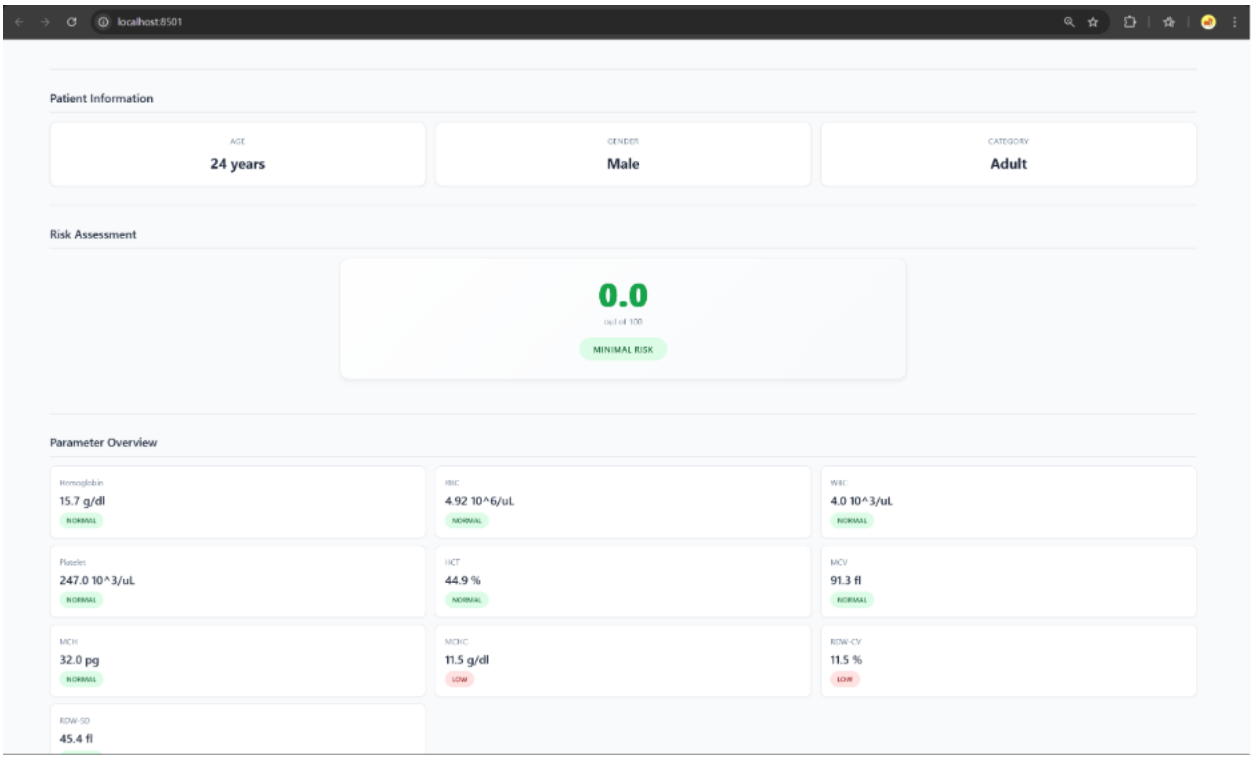
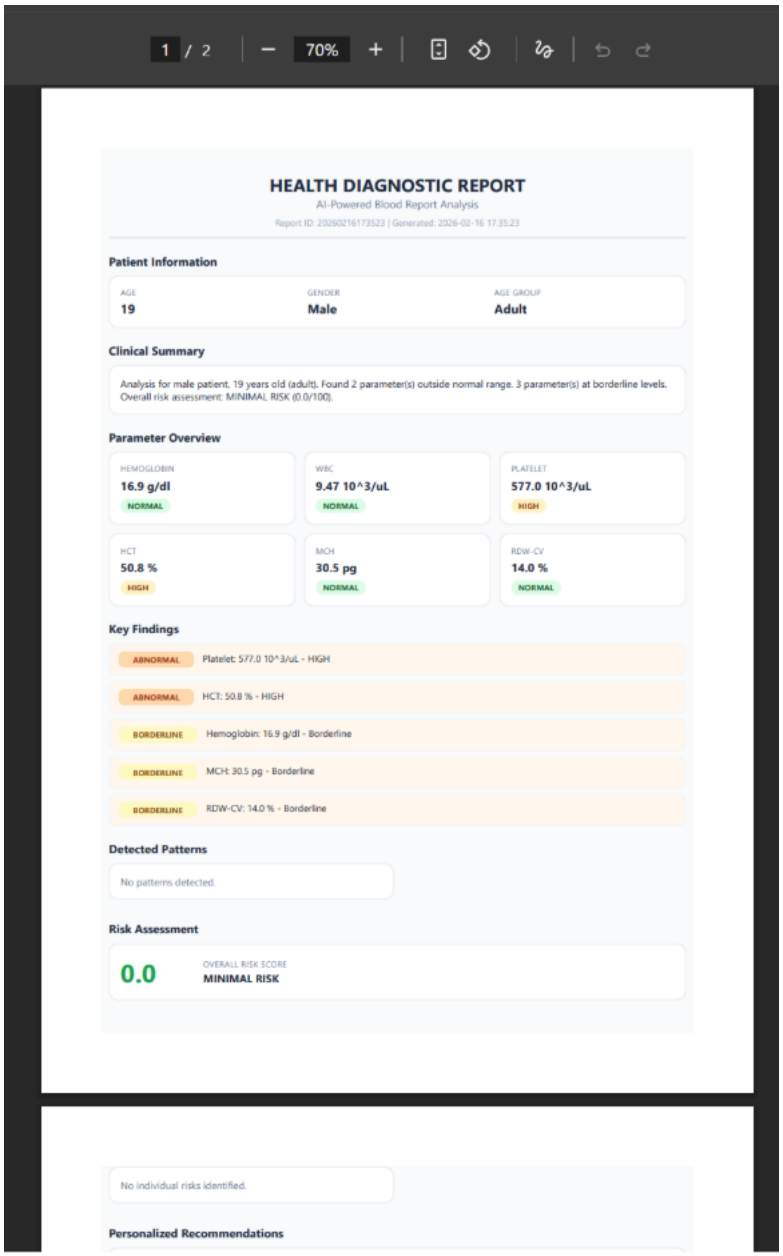

15. User Interface

A Streamlit-based web interface allows users to:

- Upload reports
- View real-time interpretation
- Visualize findings
- Download structured output reports

This enables accessibility without requiring complex frontend development.


localhost:8501
Patient Information
AGE
24 years
GENDER
Male
CATEGORY
Adult
Risk Assessment
0.0
out of 100
MINIMAL RISK
Parameter Overview
Hemoglobin
15.7 g/dl
NORMAL
RBC
4.92 10^6/uL
NORMAL
WBC
4.0 10^3/uL
NORMAL
Platelet
247.0 10^3/uL
NORMAL
HCT
44.9 %
NORMAL
MCV
91.3 fl
NORMAL
MCH
32.0 pg
NORMAL
MCHC
11.5 g/dl
LOW
RDW-CV
11.5 %
LOW
RDW-SD
45.4 fl



1 / 2 70%

HEALTH DIAGNOSTIC REPORT

AI-Powered Blood Report Analysis

Report ID: 20260216173523 | Generated: 2026-02-16 17:35:23

Patient Information

AGE	GENDER	AGE GROUP
19	Male	Adult

Clinical Summary

Analysis for male patient, 19 years old (adult). Found 2 parameter(s) outside normal range. 3 parameter(s) at borderline levels. Overall risk assessment: MINIMAL RISK (0.0/100).

Parameter Overview

HEMOGLOBIN	WBC	PLATELET
16.9 g/dl NORMAL	9.47 10^3/uL NORMAL	577.0 10^3/uL HIGH
HCT	MCH	RDW-CV
50.8 % HIGH	30.5 pg NORMAL	14.0 % NORMAL

Key Findings

- ABNORMAL Platelet: 577.0 10^3/uL - HIGH
- ABNORMAL HCT: 50.8 % - HIGH
- BORDERLINE Hemoglobin: 16.9 g/dl - Borderline
- BORDERLINE MCH: 30.5 pg - Borderline
- BORDERLINE RDW-CV: 14.0 % - Borderline

Detected Patterns

No patterns detected.

Risk Assessment

0.0 OVERALL RISK SCORE MINIMAL RISK

No individual risks identified.

Personalized Recommendations

16. System Outcomes & Contributions

Through this project, the system achieves:

- Modular multi-model AI architecture
- Structured clinical parameter interpretation
- Rule-based pattern detection with confidence scoring

- Demographic contextual reasoning
- Clinical-style narrative synthesis
- Personalized recommendation generation
- Fully functional end-to-end health diagnostics pipeline
- Error-resilient workflow orchestration

17. Technical Strengths

- Multi-format ingestion
- OCR-enabled extraction
- Regex-based structured parsing
- Configuration-driven reference management
- Rule-based explainable AI logic
- Correlation-based reasoning
- Domain-weighted risk scoring
- Modular architecture
- Streamlit-based interface
- Fault-tolerant orchestration